INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Electronics & Communication Engineering

Semester – First

End-Term Examination December– 2025

Subject Name/ Code: - FOCP CS-1003

Max. Marks: 60
Time Allotted: 3Hrs

NOTE: 1. All questions are compulsory.
2. Mark answers number clearly.

S. No.	Question	Marks
1.	**a)** Let the representation of a number in base 3 be 210. What is the hexadecimal representation of the number? **b)** If x and y are two decimal digits and $(0.1101)_2 = (0.8xy5)_{10}$, then what will be the decimal value of x + y? **c)** Consider the equation $(43)_x = (y3)_8$ where x and y are unknown. Find the value of x and y. **d)** A particular number is written as 132 in radix-4 representation. What will be its equivalent number in radix-5 representation?	2+3+3+2 = 10

2.	**a)** Evaluate the expression and find x,y,z?
	int x = 5, y = 10;
	int *p = &x, *q = &y;
	int z = (*p)++ + ++(*q) + (x += 3) + (y -= 2);
	printf("%d %d %d", x, y, z);
	b) Write the algorithm and draw the flowchart to find all the roots of a quadratic equation $ax^2+bx+c=0$
	c) Write a C program to print the following pattern
	A
	A B C
	A B C D E
	A B C D E F G
	A B C D E F G H I
		4+3+3=10

3	**a)** In a small firm employee numbers are given in serial numerical order, that is 1, 2, 3, etc. **Write a C program to implement the below operations.** i. Create a file of employee data with following information: employee number, name, sex, gross salary. ii. If more employees join, append their data to the file. iii. If an employee with serial number 25 (say) leaves, delete the record by making gross salary 0. iv. If some employee's gross salary increases, retrieve the record and update the salary. **b)** Write a C program to rotate an array with 180^0 anticlockwise?	5+5=10

4. **a)** Find the output of the following
(Mention all solutions at a single place) **(1 mark each)** **4+6=10**

i)

```
#include<stdio.h>
void mystery(int *ptra, int *ptrb)
{
  int *temp;
  temp = ptrb;
  ptrb = ptra;
  ptra = temp;
}
int main()
{
  int a=2016, b=0, c=4, d=42;
  mystery(&a, &b);
  if (a < c)
    mystery(&c, &a);
  mystery(&a, &d);
  printf("%d", a);
}
```

ii)

```
#include<stdio.h>
int main()
{
  int array[] = {3, 5, 1, 4, 6, 2};
  int done = 0;
  int i;

  while (done == 0)
  {
    done = 1;
    for (i = 0; i <= 4; i++)
    {
      if (array[i] < array[i+1])
      {
        swap(&array[i],
&array[i+1]);
        done = 0;
      }
    }
    for (i = 5; i >= 1; i--)
    {
      if (array[i] > array[i-1])
      {
        swap(&array[i],    &array[i-
1]);
        done = 0;
      }
    }
  }

  printf("%d", array[3]);
}
```

iii)

```
#include<stdio.h>
int main()
{
  int a = 300;
  char *b = (char *)&a;
  *++b = 2;
  printf("%d ",a);
  return 0;
}
```

iv)

```
#include<stdio.h>
int main()
{
  int array[5][5];
  printf("%d",( (array == *array) &&
(*array == array[0]) ));
  return 0;
}
```

b) Find the error in each of the following program segments and correct the error. (Mention all solutions at a single place) **(2 marks each)**

i)
```
#define SIZE = 100;
```

ii)
```
# include <stdio.h>
# include <string.h>
int main( )
{
char *str1 = "United" ;
char *str2 = "Front" ;
char *str3 ;
str3 = strcat ( str1, str2 ) ;
printf ( "%s\n", str3 ) ;
return 0 ;
}
```

iii)
```
# include <stdio.h>
int main( )
{
 int three[ 3 ][ ] = {
        2, 4, 3,
        6, 8, 2,
        2, 3, 1
        } ;
 printf ( "%d\n", three[ 1 ][ 1 ] ) ;
 return 0 ;
}
```

5.

a) Write a C program to find prime numbers?

b) Consider the following situations in a C program:

- A function returns the address of its local variable to the caller.
- A dynamically allocated block using malloc () is freed, but the program continues to use the same pointer to access data.
- A pointer to a structure element is stored, but later the structure variable goes out of scope.

i) What common pointer problem is occurring in all three situations?

ii) Explain why this problem is dangerous.

5+5 = 10

6.

a) A "C program" dynamically allocates memory as follows:

```
int *p, *q;
p = (int *) malloc (5 *sizeof(int));
q = p + 3;
*(q - 1) = 10;
*q = 20;
*(p + 4) = *(q - 1) + *q;
```

Assume that the Base address returned by malloc = 2000 and sizeof(int) = 4 bytes

1. Compute the address accessed by each pointer expression: q – 1, q, p+4;
2. List the final contents of the allocated 5-element memory block.

4+2+2+2=10

Solve the following with proper steps:

b) Find the address of Q[2][5] for char Q8][7] in column-major order, base = 2500.

c) Find the address of R[5][1][6] for long long R[9][2][9] in column-major order, base = 10000.

d) Find the address of K[10] for double K[50], base = 3000.

*********************************Best of Luck*********************************